Class Equation, Conjugacy, p-Groups and Finite Abelian Groups

From the class book: Chapter 11 (Sections 11.1 to 11.4), Chapter 12 (Sections 12.1 to 12.3), Chapter 13 (Section 13.1)

1. Key definitions

External direct product. For groups $G_1, \dots, G_n$, the set $G_1 \times \cdots \times G_n$ with componentwise operation $(a_i)(b_i) = (a_i b_i)$. It has order $|G_1| \cdots |G_n|$.

Internal direct product. G is the internal direct product of subgroups H, K if $H \trianglelefteq G$, $K \trianglelefteq G$, $H \cap K = \{e\}$ and $G = HK$. Then $G \cong H \times K$, every g is uniquely hk, and $H \cong G/K$.

Conjugacy class. b is conjugate to a if $b = xax^{-1}$ for some $x \in G$. Conjugacy is an equivalence relation; the class of a is $\mathrm{cl}(a) = \{xax^{-1} : x \in G\}$. $\mathrm{cl}(a) = \{a\} \iff a \in Z(G)$.

Centralizer and normalizer. $C(a) = \{x \in G : xa = ax\}$; $N(H) = \{x \in G : xHx^{-1} = H\}$.

p-group; Sylow p-subgroup. A group in which every element has order a power of p. A finite group is a p-group iff $|G| = p^n$. If $|G| = p^a m$, $p \nmid m$, a subgroup of order p^a is a Sylow p-subgroup.

2. Main results

Theorem (orders in direct products)
$o(a_1, \dots, a_n) = \mathrm{lcm}(o(a_1), \dots, o(a_n))$. $G_1 \times G_2$ is abelian iff both factors are. $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ (cyclic) iff $\gcd(m, n) = 1$; $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}$ is cyclic iff the n_i are pairwise coprime.

Theorem (structure of U(n))
If $\gcd(m, n) = 1$, $U(mn) \cong U(m) \times U(n)$. For an odd prime p, $U(p^k) \cong \mathbb{Z}_{p^k - p^{k-1}}$; $U(2)$, $U(4)$ are cyclic and $U(2^k) \cong \mathbb{Z}_2 \times \mathbb{Z}_{2^{k-2}}$ for $k \geq 3$. Hence $U(n)$ is cyclic iff $n = 1, 2, 4, p^k, 2p^k$ (p odd).

Theorem (Fundamental Theorem of Finite Abelian Groups)
Every finite abelian group is, uniquely up to order of factors, a product of cyclic groups of prime-power order (elementary divisors): $G \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}}$. Equivalently $G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}$ with $n_1 \mid n_2 \mid \cdots \mid n_r$ (invariant factors), uniquely.

Consequences: if $n = p_1^{a_1} \cdots p_k^{a_k}$, the number of abelian groups of order n is $p(a_1) \cdots p(a_k)$ (p = partition function: $p(1..7) = 1, 2, 3, 5, 7, 11, 15$). The converse of Lagrange holds for finite abelian groups. In $\prod \mathbb{Z}_{n_i}$ the number of elements of order dividing m is $\prod \gcd(m, n_i)$.

Theorem (size of a class)
$|\mathrm{cl}(a)| = [G : C(a)] = |G|/|C(a)|$. So every class size divides $|G|$. Conjugate elements have equal orders (the converse is false). The number of conjugates of a subgroup H is $[G : N(H)]$; H is normal iff it has exactly one conjugate.

Theorem (Class Equation)
For finite G,

$$|G| = |Z(G)| + \sum_{a} [G : C(a)],$$

the sum over one representative of each class with more than one element.

Corollary (p-groups)
If $|G| = p^n$, $n \geq 1$, then $Z(G) \neq \{e\}$ (every non-central term of the class equation is divisible by p). Every group of order p^2 is abelian ($\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$). A non-abelian group of order p^3 has $|Z(G)| = p$, $G/Z(G) \cong \mathbb{Z}_p \times \mathbb{Z}_p$, every non-central class of size p, and $p^2 + p - 1$ classes:
$p^3 = \underbrace{1 + \cdots + 1}_{p} + \underbrace{p + \cdots + p}_{p^2 - 1}$.

Proof idea for order p^2: if G were non-abelian, $|Z(G)| = p$ and $G/Z(G)$ would be cyclic of order p, which forces G abelian.

Theorem (Cauchy)
If a prime p divides $|G|$, then G has an element (hence a subgroup) of order p.

Theorem (conjugacy in S_n and A_n)
Two permutations are conjugate in S_n iff they have the same cycle type, so S_n has $p(n)$ classes. $\sigma(a_1\, a_2 \cdots a_r)\sigma^{-1} = (\sigma(a_1)\, \sigma(a_2) \cdots \sigma(a_r))$. With a_k cycles of length k (fixed points counted as 1-cycles),

$$|\text{class}| = \frac{n!}{\prod_k k^{a_k}\, a_k!}, \qquad |C(\sigma)| = \prod_k k^{a_k}\, a_k!.$$

An S_n-class lying in A_n splits into two A_n-classes of equal size iff its cycle lengths are distinct and odd; otherwise it stays one class.

Commuting probability: if $k(G)$ is the number of classes, $P(xy = yx) = k(G)/|G|$; for non-abelian G this is at most $5/8$ (attained by Q_8, D_4).

3. Standard tables

Class equations

Group	Class equation	Classes
abelian, order n	$1+1+\cdots+1$	n
S_3	$6=1+2+3$	3
$Q_8,\ D_4$	$8=1+1+2+2+2$	5
D_5 (order 10)	$10=1+2+2+5$	4
D_6 (order 12)	$12=1+1+2+2+3+3$	6
A_4	$12=1+3+4+4$	4
S_4	$24=1+3+6+6+8$	5
A_5	$60=1+12+12+15+20$	5
S_5	$120=1+10+15+20+20+24+30$	7
A_6	$360=1+40+40+45+72+72+90$	7
S_6	$720=1+15+15+40+40+45+90+90+120+120+144$	11

Dihedral group D_n of order $2n$: for n odd, $2n=1+\underbrace{2+\cdots+2}_{(n-1)/2}+n$, with $(n+3)/2$ classes; for n even, $|Z(D_n)|=2$ and there are $(n+6)/2$ classes.

U(n) and abelian groups of prime-power order

n	$\lvert U(n)\rvert$	$U(n)\cong$	cyclic?	max order
8, 12	4	$\mathbb{Z}_2\times\mathbb{Z}_2$	no	2
9, 18	6	$\mathbb{Z}_6$	yes	6
15, 16, 20	8	$\mathbb{Z}_2\times\mathbb{Z}_4$	no	4
40	16	$\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_4$	no	4

Order	Abelian groups	Elements of order p
p^2	$\mathbb{Z}_{p^2},\ \mathbb{Z}_p^2$	$p-1,\ p^2-1$
p^3	$\mathbb{Z}_{p^3},\ \mathbb{Z}_{p^2}\times\mathbb{Z}_p,\ \mathbb{Z}_p^3$	$p-1,\ p^2-1,\ p^3-1$
16	$\mathbb{Z}_{16},\ \mathbb{Z}_8\times\mathbb{Z}_2,\ \mathbb{Z}_4\times\mathbb{Z}_4,\ \mathbb{Z}_4\times\mathbb{Z}_2^2,\ \mathbb{Z}_2^4$	1, 3, 3, 7, 15

In $\mathbb{Z}_{p^{a_1}}\times\cdots\times\mathbb{Z}_{p^{a_r}}$ exactly p^r elements satisfy $x^p=e$. Counts: order 32: $p(5)=7$; order 72: $3\cdot 2=6$; order 1000: $3\cdot 3=9$.

4. Exam shortcuts and standard counter-examples

> **Exam shortcut.** Invariant factors from elementary divisors: list prime powers for each prime in decreasing order; multiply the largest of each prime to get n_r, the next largest to get n_{r-1}, and so on. Use $\mathbb{Z}_m\times\mathbb{Z}_n\cong\mathbb{Z}_{\mathrm{lcm}(m,n)}\times\mathbb{Z}_{\gcd(m,n)}$; e.g. $\mathbb{Z}_{12}\times\mathbb{Z}_{18}\cong\mathbb{Z}_6\times\mathbb{Z}_{36}$.

> **Exam shortcut.** An abelian group of order n is cyclic iff for every prime $p\mid n$ it has exactly $p-1$ elements of order p. Cyclic subgroups of order d = (elements of order d)/$\varphi(d)$.

Exam shortcut. Testing a proposed class equation: the 1's give $|Z(G)|$, which must divide $|G|$ and every class size; each class size divides $|G|$; for $a \notin Z(G)$, $|C(a)|$ is a proper multiple of $|Z(G)|$; $G/Z(G)$ is never cyclic of order > 1; a p-group has $p \mid |Z(G)|$.

Standard counter-examples. $\mathbb{Z}_2 \times \mathbb{Z}_4 \ncong \mathbb{Z}_8$; $\mathbb{Z}_2 \times \mathbb{Z}_2 \ncong \mathbb{Z}_4$. Q_8 and D_4 (also $\mathbb{Z}_4$ and K_4) have the same class equation but are not isomorphic. Elements of the same order need not be conjugate (in D_4, r^2 and a reflection both have order 2). $(1\,2\,3)$ and $(1\,3\,2)$ are conjugate in S_4 but not in A_4. $\mathbb{Z} \times \mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z}_m$ ($m > 1$) are not cyclic. Standard internal products: $\mathbb{R}^* = \mathbb{R}^+ \times \{\pm 1\}$, $\mathbb{C}^* = \mathbb{R}^+ \times S^1$, $D_6 \cong S_3 \times \mathbb{Z}_2$.

Exam shortcut. A finite group with $a^2 = e$ for all a is $\mathbb{Z}_2^n$, so its order is 2^n. Number of classes of S_n for $n = 3, \dots, 7$: 3, 5, 7, 11, 15.

5. Solved examples

Example 1. Find the number of elements of order 4 and the number of cyclic subgroups of order 4 in $\mathbb{Z}_4 \times \mathbb{Z}_8$.

Solution. Elements with order dividing 4: $\gcd(4,4)\gcd(4,8) = 16$. Order dividing 2: $\gcd(2,4)\gcd(2,8) = 4$. So exactly $16 - 4 = 12$ elements have order 4. Each cyclic subgroup of order 4 contains $\varphi(4) = 2$ generators, so there are $12/2 = 6$ such subgroups.

Example 2. Determine $U(105)$ up to isomorphism, the largest order of an element, and the number of elements of order 2.

Solution. $105 = 3 \cdot 5 \cdot 7$, so $U(105) \cong U(3) \times U(5) \times U(7) \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{12}$ (invariant factors), of order 48. The largest element order is $\operatorname{lcm}(2,4,6) = 12$. Elements with $x^2 = e$: $2 \cdot 2 \cdot 2 = 8$, so 7 elements of order 2. In particular $U(105)$ is not cyclic.

Example 3. How many abelian groups of order 144 are there, and which of them is cyclic? Write $\mathbb{Z}_4 \times \mathbb{Z}_6 \times \mathbb{Z}_{18}$ in invariant factor form.

Solution. $144 = 2^4 \cdot 3^2$: $p(4)p(2) = 5 \cdot 2 = 10$ groups; only $\mathbb{Z}_{144}$ is cyclic. For the second part, $\mathbb{Z}_4 \times \mathbb{Z}_6 \times \mathbb{Z}_{18} \cong \mathbb{Z}_4 \times (\mathbb{Z}_2 \times \mathbb{Z}_3) \times (\mathbb{Z}_2 \times \mathbb{Z}_9)$. Prime powers: 2-part $4, 2, 2$; 3-part $9, 3$. Then $n_3 = 4 \cdot 9 = 36$, $n_2 = 2 \cdot 3 = 6$, $n_1 = 2$: the group is $\mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{36}$ (order 432 as expected).

Example 4. In S_7, find the number of conjugates of $\sigma = (1\,2)(3\,4\,5)$ and $|C(\sigma)|$. Also compute $\tau(1\,3\,6)\tau^{-1}$ for $\tau = (1\,2\,6)(4\,5)$.

Solution. Cycle type: one 2-cycle, one 3-cycle, two fixed points. $|C(\sigma)| = 2 \cdot 3 \cdot 1^2 \cdot 2! = 12$ and $|\mathrm{cl}(\sigma)| = 5040/12 = 420$. For the conjugate, relabel: $\tau(1) = 2,\ \tau(3) = 3,\ \tau(6) = 1$, so $\tau(1\,3\,6)\tau^{-1} = (2\,3\,1) = (1\,2\,3)$.

Example 5. Which can be class equations: (a) $8 = 1 + 1 + 2 + 4$, (b) $10 = 1 + 1 + 2 + 2 + 2 + 2$, (c) $21 = 1 + 3 + 3 + 7 + 7$?

Solution. (a) No: $|Z(G)| = 2$; a class of size 4 means $|C(a)| = 2 = |Z(G)|$, but $C(a)$ contains $Z(G)$ and $a \notin Z(G)$, so $|C(a)| > 2$. (b) No: $|Z(G)| = 2$ gives $G/Z(G)$ of order 5, cyclic, so G is abelian, contradicting non-trivial classes. (c) Yes: the non-abelian group $\mathbb{Z}_7 \rtimes \mathbb{Z}_3$ (exists since $3 \mid 7 - 1$) has trivial centre; its 6 elements of order 7 form two classes of 3 (centralizer = the Sylow 7-subgroup), and its 14 elements of order 3 form two classes of 7. It has 5 classes.

Example 6. Let G be non-abelian of order 27. Find $|Z(G)|$, the number of conjugacy classes, and the probability that two random elements commute.

Solution. For order p^3, $p=3$: $|Z(G)| = 3$, every non-central class has 3 elements, $27 = 3\cdot 1 + 8\cdot 3$, so $p^2 + p - 1 = 11$ classes. Probability $= k(G)/|G| = 11/27$.

Example 7. In A_5, into how many classes do the 5-cycles fall, and what is $|C_{A_5}((1\,2\,3\,4\,5))|$?

Solution. In S_5 the 5-cycles form one class of $120/5 = 24$ elements. The cycle lengths (just 5) are distinct and odd, so in A_5 the class splits into two classes of 12. Hence $|C_{A_5}(\sigma)| = 60/12 = 5$, i.e. $C_{A_5}(\sigma) = \langle \sigma \rangle$.